

Prescribed Isolated Intervals in Higher Fold Sumsets

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Abstract

We give an explicit construction of a finite set of integers whose h -fold sumset has any prescribed sufficiently separated family of nontrivial intervals and no others. This answers Problems 9 and 10 in [Nathanson's problem list](#) with an explicit set of size $2q(n+1)$, and also answers the higher-order Problem 11. We then replace the initial exponential labels by finite moment labels. For fixed h , this gives polynomial diameter in the number of prescribed target points and works for an arbitrary finite target set. Both label choices are instances of a common separated-label principle. For double sumsets, a Sidon-label instance improves the diameter bound to quadratic. We record a small counterexample to the strict inequality proposed in Problem 7(2).

1 A short separation lemma

For an integer vector $v = (v_0, \dots, v_{s-1})$, write $\|v\|_1 = \sum_r |v_r|$.

Lemma 1 (Positional separation). *Let $B, M \geq 1$ and $a_r = MB^r$ for $0 \leq r < s$. If $\lambda, \mu \in \mathbb{Z}^s$, $\lambda \neq \mu$, and $\|\lambda\|_1 + \|\mu\|_1 \leq B$, then*

$$\left| \sum_{r=0}^{s-1} (\lambda_r - \mu_r) a_r \right| \geq M.$$

Proof. Put $z = \lambda - \mu$ and let R be the largest index with $z_R \neq 0$. We have $\|z\|_1 \leq B$. If $R = 0$, the result is immediate. If $R \geq 1$, then

$$\begin{aligned} \left| \sum_{r=0}^{s-1} z_r a_r \right| &\geq M \left(|z_R| B^R - \sum_{r < R} |z_r| B^r \right) \\ &\geq M (B^R - (B-1)B^{R-1}) \geq M. \end{aligned}$$

Here the sum of the absolute values of the lower coefficients is at most $B - |z_R| \leq B - 1$. □

Proposition 1 (Separated-label principle). *Let $h \geq 2$, $d \geq 1$, and let $S = \{t_0, \dots, t_{m-1}\} \subseteq \mathbb{Z}$ be nonempty. Put*

$$H = h(h-1), \quad T = \max_{t \in S} |t|.$$

Suppose that $M \geq 2hT + d$ and that integers $a_0, \dots, a_{m-1}$ satisfy

$$\left| \sum_{r=0}^{m-1} (\lambda_r - \mu_r) a_r \right| \geq M \tag{1}$$

whenever $\lambda, \mu \in \mathbb{Z}^m$ are distinct and

$$\|\lambda\|_1 + \|\mu\|_1 \leq 2H.$$

Then

$$A = \{t_r + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}$$

has $|A| = 2m$ and

$$hA = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

Proof. Attach to $t_r + (h-1)a_r$ the label vector $(h-1)e_r$ and offset t_r , and attach to $-a_r$ the label vector $-e_r$ and offset 0. Distinct tagged generators have distinct label vectors, combined norm at most $2H$, and offsets differing by at most $2T < M$. Thus (1) shows that their integer values are distinct, so $|A| = 2m$.

Every element of hA is a bounded offset plus $\sum_r v_r a_r$, where $\|v\|_1 \leq H$. Let p_r and u_r count the positive and negative terms with index r in an h -term sum. A zero label vector satisfies $(h-1)p_r = u_r$ for every r . Since

$$\sum_r (p_r + u_r) = h \sum_r p_r = h,$$

there is exactly one positive term and $h-1$ matching negative terms. Hence the zero-label sums are exactly

$$(t_r + (h-1)a_r) + (h-1)(-a_r) = t_r.$$

Every other sum has a nonzero label vector. Such a vector determines its multiset of terms: if two multisets with counts (p_r, u_r) and (p'_r, u'_r) had the same nonzero vector but were different, then

$$u_r - u'_r = (h-1)(p_r - p'_r).$$

Summing and using that both multisets have h terms shows that $\sum_r p_r = \sum_r p'_r$. At an index with $p_r > p'_r$, the first multiset would contain at least one positive term and $h-1$ matching negative terms. It would therefore be the zero-label matched sum, a contradiction.

Finally, every representation offset has absolute value at most hT . Distinct exceptional sums have distinct label vectors, and an exceptional sum also has a label vector distinct from the zero vector of a target. Equation (1) and the offset bound leave a gap of at least $M - 2hT \geq d$. $\square$

2 The construction

Theorem 1 (Prescribed isolated intervals). *Let $h \geq 2$, $n \geq 1$, $d \geq 2$, $q \geq 1$, and*

$$C = \{c_1 < \dots < c_q\} \subseteq \mathbb{Z}$$

with $c_{i+1} - c_i \geq n + 2$ for all i . There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2q(n+1)$ such that

$$hA = S \dot{\cup} E, \quad S = \bigcup_{i=1}^q [c_i, c_i + n],$$

where

$$|x - y| \geq d$$

whenever $x, y \in E$ are distinct or $x \in E$ and $y \in S$. Consequently, the maximal intervals in hA of length at least one are exactly the intervals $[c_i, c_i + n]$.

Proof. List the $q(n+1)$ elements of S as $t_0, \dots, t_{m-1}$, where $m = q(n+1)$. Set

$$T = \max_{t \in S} |t|, \quad M = 2hT + d, \quad B = 2h(h-1), \quad a_r = MB^r,$$

and define

$$A = \{t_r + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}.$$

Lemma 1 verifies (1) with $B = 2h(h-1)$. Proposition 1 gives $|A| = 2m$, the decomposition $hA = S \dot{\cup} E$, and the required distance bounds. Since $d \geq 2$ and the intervals in S are separated by at least one missing integer, no exceptional point can extend a target interval or form a new nontrivial interval. $\square$

Corollary 1 (Single interval replacement). *For all $h \geq 2$, $c \in \mathbb{Z}$, $n \geq 1$, and $d \geq 2$, there is a set $A \subseteq \mathbb{Z}$ with $|A| = 2n+2$ such that $[c, c+n] \subseteq hA$, every element of $hA \setminus [c, c+n]$ is at distance at least d from every other such element and from $[c, c+n]$, and the longest interval in hA has length n .*

Proof. Apply Theorem 1 with $C = \{c\}$. $\square$

Corollary 2 (Problems 9–11). *In the notation of Problems 9–11 of Nathanson’s problem list, the assertions hold for every N_q allowed in the question, with*

$$k_C = 2q(n+1).$$

Proof. The hypotheses give $c_{i+1} - c_i \geq N_q \geq n+2$. Apply Theorem 1 with $d = 2$, taking $h = 2$ for Problems 9–10 and the specified value of h for Problem 11. $\square$

Remark 1 (Diameter bound). *The construction in Theorem 1 satisfies*

$$\text{diam}(A) \leq h(2hT + d)(2h(h-1))^{q(n+1)-1} + T.$$

The bound is intentionally simple rather than optimized.

3 A polynomial diameter refinement

The power labels in Theorem 1 make the first construction transparent, but their size is exponential in the number of target points. The following finite moment encoding gives a polynomial bound for fixed h .

Lemma 2 (Moment separation). *Let $m, L \geq 1$, put*

$$R = \max(1, m-1), \quad K = LR^{L-1}, \quad Q = LK = L^2R^{L-1},$$

and define

$$b_r = \sum_{j=0}^{L-1} r^j Q^j \quad (0 \leq r < m).$$

If $z = (z_0, \dots, z_{m-1}) \in \mathbb{Z}^m$ is nonzero and $\|z\|_1 \leq L$, then

$$\left| \sum_{r=0}^{m-1} z_r b_r \right| \geq 1.$$

Proof. For $0 \leq j < L$, put

$$\mu_j = \sum_{r=0}^{m-1} z_r r^j.$$

Not all μ_j vanish. Indeed, if the support of z has size s , then $s \leq L$, and the first s equations $\mu_j = 0$ would form an invertible Vandermonde system on the distinct support indices.

Since $|\mu_j| \leq LR^{L-1} = K$, the moment vector has ℓ^1 norm at most $LK = Q$. A nonzero integer coefficient vector with ℓ^1 norm at most Q cannot evaluate to zero in base Q : if J is its largest nonzero index, the leading term has absolute value at least Q^J , while all lower terms together have absolute value at most $(Q-1)Q^{J-1}$ when $J \geq 1$; the case $J = 0$ is immediate. Applying this positional domination to $(\mu_0, \dots, \mu_{L-1})$ proves the claim. $\square$

Theorem 2 (Polynomial diameter realization). *Let $h \geq 2$, $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$hA = S \dot{\cup} E,$$

where

$$|x - y| \geq d$$

whenever $x, y \in E$ are distinct or $x \in E$ and $y \in S$.

More precisely, let $\tau \in h\mathbb{Z}$ minimize

$$T = \max_{t \in S} |t - \tau|.$$

Put

$$L = 2h(h-1), \quad R = \max(1, m-1), \quad Q = L^2 R^{L-1},$$

and

$$P_h(m) = \sum_{j=0}^{L-1} (m-1)^j Q^j.$$

Then A can be chosen with

$$\text{diam}(A) \leq h(2hT + d)P_h(m) + T.$$

For fixed h , the factor $P_h(m)$ is bounded by a polynomial in m of degree $L(L-1)$.

Proof. Enumerate $S = \{t_0, \dots, t_{m-1}\}$ and use the parameters in Lemma 2. Set

$$M = 2hT + d, \quad a_r = M \sum_{j=0}^{L-1} r^j Q^j,$$

and first construct the centered set

$$A_0 = \{t_r - \tau + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}.$$

Finally, put

$$A = A_0 + \tau/h.$$

If $\lambda \neq \mu$ and $\|\lambda\|_1 + \|\mu\|_1 \leq L$, then $z = \lambda - \mu$ is nonzero and $\|z\|_1 \leq L$. Lemma 2, after multiplication by M , therefore verifies (1) for the centered target set. Proposition 1 gives $|A_0| = 2m$, the matching-label decomposition, and the required distance bounds. Translating A_0 by τ/h translates hA_0 by τ and does not change its diameter.

The labels a_r are strictly increasing. Every element of A_0 lies between $-a_{m-1}$ and $T + (h-1)a_{m-1}$. Consequently,

$$\text{diam}(A) = \text{diam}(A_0) \leq ha_{m-1} + T = h(2hT + d)P_h(m) + T.$$

For $m \geq 2$, the displayed expression for $P_h(m)$ is a polynomial of degree $L(L-1)$; the case $m = 1$ is immediate. $\square$

Corollary 3 (Translation-invariant interval bound). *Under the hypotheses of Theorem 1, let*

$$m = q(n + 1), \quad D = c_q + n - c_1.$$

There is a set A with the same isolated interval pattern and $|A| = 2q(n + 1)$ whose diameter is polynomial in m for fixed h , with

$$T \leq \left\lceil \frac{D + h}{2} \right\rceil$$

in the bound of Theorem 2. In particular, the bound is independent of the absolute location of the prescribed intervals.

Proof. Apply Theorem 2 to $S = \bigcup_i [c_i, c_i + n]$. A multiple τ of h can be chosen within $h/2$ of the midpoint of $[c_1, c_q + n]$, giving the stated bound for T . $\square$

Theorem 3 (Quadratic diameter for double sumsets). *Let $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$2A = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

More precisely, let $\tau \in 2\mathbb{Z}$ minimize

$$T = \max_{t \in S} |t - \tau|,$$

and let $p \geq m$ be an odd prime. For $0 \leq r < m$, let ρ_r be the least nonnegative residue of r^2 modulo p , and put

$$b_r = 2p^2 + 2pr + \rho_r.$$

Then A can be chosen with

$$\text{diam}(A) \leq 8(4T + d)p^2 + T.$$

For $m \geq 2$, Bertrand's postulate allows $p < 2m$, giving a quadratic bound in m . Since p is an integer, the resulting explicit estimate is

$$\text{diam}(A) \leq 8(4T + d)(2m - 1)^2 + T.$$

Proof. First observe that $\{b_0, \dots, b_{m-1}\}$ is a Sidon set, with repetitions allowed in pair sums. Indeed, suppose

$$b_i + b_j = b_k + b_\ell.$$

Reducing modulo $2p$ gives $\rho_i + \rho_j = \rho_k + \rho_\ell$, because both sides lie between 0 and $2p - 2$. It follows that $i + j = k + \ell$. Reducing the residue identity modulo p also gives

$$i^2 + j^2 \equiv k^2 + \ell^2 \pmod{p}.$$

Since p is odd, we obtain $ij \equiv k\ell \pmod{p}$. Thus the two pairs are the roots of the same quadratic polynomial over $\mathbb{F}_p$. The unordered pairs $\{i, j\}$ and $\{k, \ell\}$ agree modulo p , hence agree as integer pairs because all indices lie in $[0, p)$.

Moreover,

$$2p^2 \leq b_r < 4p^2.$$

Consequently, an equality between two sums of the b_r involving at most four terms in total must have the same number of terms on both sides: one term is strictly smaller than any sum of two terms. Equal one-term sums are trivial, and equal two-term sums are trivial by the Sidon property. Therefore

$$\sum_r z_r b_r \neq 0$$

for every nonzero integer vector z with $\|z\|_1 \leq 4$.

Set $M = 4T + d$, enumerate $S = \{t_0, \dots, t_{m-1}\}$, and apply Proposition 1 to the centered targets $t_r - \tau$ and the labels $a_r = Mb_r$. Finally translate the resulting basis by $\tau/2$. Since $b_r < 4p^2$, its diameter is at most

$$2M \max_r b_r + T \leq 8(4T + d)p^2 + T.$$

The same construction has quadratic diameter from below. Indeed, translation does not change diameter, the negative generators contain $-Mb_0$ and $-Mb_{m-1}$, and

$$b_{m-1} - b_0 = 2p(m-1) + \rho_{m-1} \geq 2m(m-1).$$

Consequently,

$$\text{diam}(A) \geq 2Mm(m-1).$$

□

Proposition 2 (A Sidon-label diameter barrier). *Let $B = \{b_1, \dots, b_m\} \subseteq \mathbb{Z}$ be a Sidon set, where repetitions are allowed in pair sums. Then*

$$\text{diam}(B) \geq \binom{m}{2}.$$

Consequently, any tagged construction that contains the generators $-Mb_i$ has diameter at least $M\binom{m}{2}$.

Proof. Order the labels so that $b_1 < \dots < b_m$. The $\binom{m}{2}$ positive differences $b_j - b_i$, with $i < j$, are distinct. Indeed, an equality $b_j - b_i = b_\ell - b_k$ gives $b_j + b_k = b_\ell + b_i$, and the Sidon property forces the two ordered index pairs to agree. Every positive difference is at most $\text{diam}(B)$, proving the first claim. The second follows because the generators $-Mb_i$ already have diameter $M \text{diam}(B)$. □

Remark 2 (Elementary lower bounds). *Let $A \subseteq \mathbb{Z}$ be finite and nonempty, put $k = |A|$, and suppose that $S \subseteq hA$ has m elements. Then*

$$m \leq \binom{k+h-1}{h}, \quad \text{diam}(A) \geq \frac{\text{diam}(S)}{h}.$$

Indeed, the first inequality counts unordered h -term multisets from A , and the second follows from

$$\text{diam}(hA) = h \text{diam}(A).$$

Thus Theorem 2 supplies a polynomial diameter bound for fixed h , while Theorem 3 gives a quadratic bound when $h = 2$. Proposition 2 shows that the quadratic order is unavoidable within the Sidon-label method. Optimizing either the cardinality or the degree for $h \geq 3$ remains open.

4 Audit notes and a small counterexample

The proof above also supplies a replacement proof for the single-interval statement in Theorem 4 of Nathanson's preprint. In the current source, the displayed separation parameter is

$$\Delta = \min(3, 2|c| + d + 2n),$$

while later estimates require a parameter larger than quantities depending on $|c|$, d , and n . The replacement argument avoids that discrepancy.

Problem 7(2) asks whether a negative element forces a strict loss in the initial interval. This is false: take $h = 2$, $k = 3$, and

$$A = \{-1, 1, 2\}.$$

Then $[0, 4] \subseteq 2A$. To see that $n_2(3) = 4$, write a nonnegative three-element basis for an initial interval as $\{0, 1, a\}$. If $a = 2$ or $a = 3$, its initial interval ends at 4, while if $a > 3$, the integer 3 is missing. Hence $\ell_2(A) = n_2(3)$ even though $\min(A) < 0$.

5 Reproducibility

The accompanying Python package constructs A , computes hA , checks all maximal consecutive runs, records the minimum exceptional spacing, and emits JSON certificates. It also includes exact small-parameter computations cross-checked against [OEIS A001208](#) and [OEIS A001209](#). The arithmetic separation, bounded-offset, matched-count, tagged-representation, exceptional-count uniqueness, generic set-level isolation, interval-enumeration, maximal-run, and exact-cardinality arguments have also been checked in Lean 4 with Mathlib. The resulting Lean theorem automatically selects the target radius and substitutes the displayed values of M and B . The finite moment-label Vandermonde argument, translation rule, closed diameter estimate, exact cardinality, midpoint-centering lemma, and interval-family theorem have also been formalized. The audit note about the earlier preprint remains an ordinary paper argument. For the shifted-Sidon refinement, Lean checks the size bounds and the reduction from order-four short-relation freedom to set-level isolation, including the explicit pair-sum argument over $\mathbb{F}_p$. It also checks translation invariance, quadratic diameter upper and lower bounds, tightened Bertrand-prime selection, exact cardinality after translation, and the final separated-interval wrapper. The midpoint translation, radius, and scale are selected automatically from an enclosing interval. Lean also checks the direct Problem 7(2) witness $\{-1, 1, 2\}$ and the containment $[0, 4] \subseteq 2A$; the classical identification $n_2(3) = 4$ remains the short paper argument given above.